LECTURES 14-15

Linear transformations of multi random variables: In this section we look at when $\varphi(x,y) = (x,y)A$, where A is a 2×2 invertible (i.e. non singular) matrix. For example, when

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

then A rotates (X,Y) through an angle θ in the clockwise direction. To find the distribution of $(U,V) = (X,Y)A$, we use the following change of variable formula. Let $\varphi = (\varphi_1,\varphi_2) : \mathbb{R}^2 \to \mathbb{R}^2$ be a map (with continuously differentiable partial derivatives) such that the Jacobian at (u,v)

$$J(\varphi_1(u,v),\varphi_2(u,v)) = \det\begin{pmatrix} \frac{\partial\varphi_1}{\partial u} & \frac{\partial\varphi_1}{\partial v} \\ \frac{\partial\varphi_2}{\partial u} & \frac{\partial\varphi_2}{\partial v} \end{pmatrix} \neq 0$$

Then the area element under the mapping $(u,v) \mapsto (x = \varphi_1(u,v), y = \varphi_2(u,v))$ makes the transformation $dudv \mapsto dxdy = |J(\varphi_1(u,v),\varphi_2(u,v))|dudv$ [1] This leads to the following change of variable formula.

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[1]Here note that infinitesemal (small) rectangle $[u,\ u+du]\times[v, v+dv]$ i.e., $dv\times dv$ is approximately mapped to the parallelogram $dx\times dy$. Now dx is the vector joining the points $(\varphi_1(u,v),\varphi_2(u,v))$ and $(\varphi_1(u+du,v),\varphi_2(u+du,v))$ and hence

$$\begin{aligned} dx &= (\varphi_1(u+du,v)-\varphi_1(u,v), \varphi_2(u+du,v)-\varphi_2(u,v)) \\ &\approx \Big(\frac{\partial\varphi_1(u,v)}{\partial u}du,\ \frac{\partial\varphi_2(u,v)}{\partial u}du\Big). \end{aligned}$$

Similarly

$$dy \approx \Big(\frac{\partial\varphi_1(u,v)}{\partial v}dv,\ \frac{\partial\varphi_2(u,v)}{\partial v}dv\Big).$$

Hence

$$\begin{aligned} dx &\approx \frac{\partial\varphi_1(u,v)}{\partial u}du\,\mathbf{i} + \frac{\partial\varphi_2(u,v)}{\partial u}du\,\mathbf{j} + 0\,\mathbf{k} \\ dy &\approx \frac{\partial\varphi_1(u,v)}{\partial v}dv\,\mathbf{i} + \frac{\partial\varphi_2(u,v)}{\partial v}dv\,\mathbf{j} + 0\,\mathbf{k}. \end{aligned}$$

$$\begin{aligned} \text{Area of the parallelogram} &= |dx\times dy| \\ &= \Big|\mathbf{k}\Big(\frac{\partial\varphi_1(u,v)}{\partial u}\frac{\partial\varphi_2(u,v)}{\partial v}dudv - \frac{\partial\varphi_2(u,v)}{\partial u}\frac{\partial\varphi_1(u,v)}{\partial v}dudv\Big)\Big| \\ &= |J(\varphi_1(u,v),\varphi_2(u,v))|\,dudv. \end{aligned}$$

Theorem 0.1 *Let D be a region in $\mathbb{R}^2$ and $f : D \to \mathbb{R}$ be continuous. Let $\varphi = (\varphi_1, \varphi_2) : O \to \mathbb{R}^2$ be such that*

(i) φ is one to one

(ii) φ_1, φ_2 have continuous paritial derivatives in O, O is an open set in $\mathbb{R}^2$

(iii) $J(\varphi_1(u,v), \varphi_2(u,v)) \neq 0$ in O

(iv) There exists $E \subseteq O$ such that E is elementary and $\varphi(E) = D$.

Then

$$\boxed{\iint_D f(x,y)dxdy = \iint_E f(\varphi_1(u,v), \varphi_2(u,v))|J(\varphi_1(u,v), \varphi_2(u,v))|dudv.}$$

As an application of the above change of variable formula, we have the following result.

Corollary 0.1 *Let (X,Y) be a random vector in $\mathbb{R}^2$ with joint pdf f and A be a non singular 2×2 matrix. Then $(U,V) = (X,Y)A$ is with pdf g given by*

$$g(u,v) \;=\; \frac{1}{|\det A|} f((u,v)A^{-1})$$

Proof: For $u, v \in \mathbb{R}$, set $R_{uv} = (-\infty, u] \times (-\infty, v]$ and $I \;=\; \varphi(R_{uv})$, where $\varphi(u,v) = (u,v)A^{-1}$. Now the distribution function $F_{(U,V)}$ of (U,V) is given by

$$\begin{aligned}
F_{(U,V)}(u,v) &= P\big((U,V) \in R_{uv}\big) \\
[\text{using } (x,y) \mapsto (x,y)A, \text{ a bijection}] &= P\big((X,Y) \in I\big) \\
&= \iint_I f(x,y)dxdy \\
(\text{Theorem 0.1 to } (u,v) \mapsto (u,v)A^{-1}) &= \iint_{R_{uv}} f((u,v)A^{-1}) \frac{1}{|\det A|} dudv.
\end{aligned}$$

Hence (U,V) has a pdf say g and is given by

$$g(u,v) \;=\; \frac{1}{|\det A|} f((u,v)A^{-1}).$$

□

Multinormal distribution. X is said to be a non degenerate multinormal with parameters $m = (m_1, m_2)$ and $\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix}$ if the joint pdf f_X is given by

$$f_X(x_1, x_2) \;=\; \frac{1}{2\pi\sqrt{\det\Sigma}} e^{-\frac{1}{2}(x-m)\Sigma^{-1}(x-m)^{\perp}},$$

where Σ is symmetric positive definite and $(x-m)^{\perp}$ is the column vector corresponding to the row vector (x_1-m_1, x_2-m_2). Also Σ is positive definite implies that $|\sigma_{12}| \leq \sigma_1\sigma_2$. Rewrite Σ as

$$\Sigma \;=\; \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

then (exercise)

$$f_X(x_1, x_2) \;=\; \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-m_1)^2}{\sigma_1^2} - \frac{2\rho(x_1-m_1)(x_2-m_2)}{\sigma_1\sigma_2} + \frac{(x_2-m_2)^2}{\sigma_2^2}\right]}.$$

Now we will see the marginal pdfs of multinormal.

Essence of the calculation is 'completing the square'. I will do 'completing the square' separately. Consider

$$\begin{aligned}
&\tfrac{(x_1-m_1)^2}{\sigma_1^2} - \tfrac{2\rho(x_1-m_1)(x_2-m_2)}{\sigma_1\sigma_2} + \tfrac{(x_2-m_2)^2}{\sigma_2^2} \\
&= \left[\left(\tfrac{(x_2-m_2)}{\sigma_2}\right)^2 - \tfrac{2\rho(x_1-m_1)(x_2-m_2)}{\sigma_1\sigma_2} + \left(\tfrac{\rho(x_1-m_1)}{\sigma_1}\right)^2\right] + (1-\rho^2)\left(\tfrac{(x_1-m_1)}{\sigma_1}\right)^2 \\
&= \left(\tfrac{(x_2-m_2)}{\sigma_2} - \tfrac{\rho(x_1-m_1)}{\sigma_1}\right)^2 + (1-\rho^2)\left(\tfrac{(x_1-m_1)}{\sigma_1}\right)^2 \\
&= \left(\tfrac{(x_2-m_2)}{\sigma_2} - a\right)^2 + (1-\rho^2)\left(\tfrac{(x_1-m_1)}{\sigma_1}\right)^2,
\end{aligned}$$

where

$$a = \frac{\rho(x_1-m_1)}{\sigma_1}.$$

Now

$$\begin{aligned}
\int_{-\infty}^{\infty} f_X(x_1,x_2)dx_2 &= \frac{e^{-\frac{1}{2}\left(\frac{(x_1-m_1)}{\sigma_1}\right)^2}}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\int_{-\infty}^{\infty} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_2-m_2)}{\sigma_2}-a\right)^2}dx_2 \\
\Big[\text{put } x=\frac{1}{\sqrt{1-\rho^2}}\Big(\frac{x_2-m_2}{\sigma_2}-a\Big)\Big] &= \frac{e^{-\frac{1}{2}\left(\frac{(x_1-m_1)}{\sigma_1}\right)^2}}{\sqrt{2\pi}\sigma_1}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}dx \\
&= \frac{1}{\sqrt{2\pi}\sigma_1}e^{-\frac{1}{2}\left(\frac{(x_1-m_1)}{\sigma_1}\right)^2}.
\end{aligned}$$

Hence $X_1 \sim N(m_1,\sigma_1)$. Similarly $X_2 \sim N(m_2,\sigma_2)$.

Theorem 0.2 *Let $X=(X_1,X_2)$ be a multinormal non degenerete random variable with parameters m and Σ. Then for any $\alpha,\beta\in\mathbb{R}$ $\alpha X_1+\beta X_2$ is a normal random variable.*

Proof: Since $X-m$ is normal with paramenters 0 and Σ, it is enough to prove the theorem when $m=0$. (exercise)

Let A be a 2×2 symmetric matrix such that $AA^{\perp}=\Sigma$ [Here the choice of A is $\Sigma^{\frac{1}{2}}$ and $\Sigma^{\frac{1}{2}}=P\Lambda^{\frac{1}{2}}P^{-1}$ where Λ is the diagonal matrix (of eigen values of Σ and $\Sigma=P\Lambda P^{-1}$) and hence $\Lambda^{\frac{1}{2}}$ is the diagonal matrix with diagonal entries as the square root of the eigen values of Λ]. Define $Y=X\Sigma^{-\frac{1}{2}}$, then using Theorem 0.1, the pdf g of Y exists and is given by

$$\begin{aligned}
g(y_1,y_2) &= |\det A| f(yA) \\
&= \sqrt{\det\Sigma}\frac{1}{2\pi\sqrt{\det\Sigma}}e^{-\frac{1}{2}yA\Sigma^{-1}Ay^{\perp}} \\
&= \frac{1}{2\pi}e^{-\frac{1}{2}\|y\|^2},\ y=(y_1,y_2).
\end{aligned}$$

Hence Y is multi normal with parameters 0 and I. Therefore $g(y_1,y_2)=g_{Y_1}(y_1)g_{Y_2}(y_2)$. This implies that Y_1 and Y_2 are independent normal random variables.

Now

$$\alpha X_1+\beta X_2 = X\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = Y\Sigma^{\frac{1}{2}}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = aY_1+bY_2$$

for some $a, b \in \mathbb{R}$ and hence is a normal random variable. (exercise)
This completes the proof. □

Remark 0.1 *The proof of Theorem 0.2 tells us some thing more. Let $X \sim N(m,\ \Sigma)$, i.e. X is a multinormal random variable with paramenters m and Σ. Set $\bar{X} = X - m$, then $\bar{X} \sim N(0, \Sigma)$.*
Then $\bar{Y} = \bar{X}\Sigma^{-\frac{1}{2}} \sim N(0, I)$. Hence

$$Y := X\Sigma^{-\frac{1}{2}} \ = \ m\Sigma^{-\frac{1}{2}} + \bar{X}\Sigma^{-\frac{1}{2}} \sim N(m\Sigma^{-\frac{1}{2}}, I).$$

Note the above is a generalization to the multidimentional case of the following result for the normal random variables.

$$X \sim N(m, \sigma^2), \text{ then } aX \sim N(a\,m, a^2\sigma^2).$$

Theorem 0.2 leads to a more general definition multinormal which includes multinormal random variables which are degenerate also.

Definition 0.1 *A random vector $X = (X_1, X_2)$ is said to be multinormal if $\alpha X_1 + \beta X_2$ is normal for all $\alpha, \beta \in \mathbb{R}$.*

Example 0.1 *Let $X_1 \sim N(0, 1)$ and $X = (X_1, X_1)$. Then any linear combination of the components of X is normally distributed. Also X doesnot have a joint density function. To see this, let $L = \{(x, y)|y = x\}$, the graph of $y = x$. Then*

$$P\{X \in L\} = P(\Omega) = 1.$$

Now suppose X has a joint pdf f, then for $L_n \ = \ L \cap [-n,\ n]$,

$$P\{X \in L_n\} \ = \ \iint_{L_n} f(x, y)dxdy = 0, \text{ for all } n \geq 1.$$

Now

$$P\{X \in L\} \ = \ \lim_{n\to\infty} P\{X \in L_n\} = 0$$

a conradiction to $P\{X \in L\} = 1$. Hence X doesn't have a density. i.e., X is an example of a degenerate multinormal distribution.

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Order Statistics. We end the discussion on transformations of random variables with order statistics. Let $X_1, X_2, \cdots, X_n$ be n random variables, the order statistics denoted by $X^{(1)}, X^{(2)}, \cdots, X^{(n)}$ of these random variables are defined as follows.

$$\begin{aligned} X^{(1)} &= \min\{X_1, X_2, \cdots, X_n\} \\ X^{(2)} &= \min\{X_j \,|\, X_j \neq X^{(1)}\} \\ &\cdots \\ &\cdots \\ X^{(n)} &= \max\{X_1, X_2, \cdots, X_n\}, \end{aligned}$$

where $[n]$ denote $\{1, 2, \cdots, n\}$.

For the rest of the section, we assume that $X_1, \cdots, X_n$ are i.i.d. random variables with distribution function of X_1 denoted by F. More over, we assume that the pdf of $(X_1, \cdots, X_n)$ exists and denote it by f. Also the distribution function F is assumed to be differentiable on $\{x : 0 < F(x) < 1\}$.

Distribution function of marginals: First we will compute the marginals of $X^{(k)}$. Let $F_{X^{(k)}}$ denote the distribution function of $X^{(k)}$. For $x \in \mathbb{R}$,

$$\begin{aligned} P\big(X^{(k)} \leq x\big) &= P\big(X_i \text{ are less than or equal to } x \text{ for atleast } k\ i's\big) \\ &= P\Big(\bigcup_{l=k}^{n} \bigcup_{\{i_1,i_2,\cdots,i_l\}\subseteq[n]} \{X_{i_1} \leq x, X_{i_2} \leq x, \cdots X_{i_l} \leq x, X_i > x, i \notin \{i_1, \cdots, i_l\}\}\Big) \\ &= \sum_{l=k}^{n} \sum_{\{i_1,i_2,\cdots,i_l\}\subseteq[n]} P\Big(X_{i_1} \leq x, \cdots X_{i_l} \leq x,\, X_i > x\, \forall\, i \notin \{i_1, \cdots i_l\}\Big) \\ &= \sum_{l=k}^{n} \binom{n}{l} P\Big(X_{i_1} \leq x, \cdots X_{i_l} \leq x,\, X_i > x\, \forall\, i \notin \{i_1, \cdots i_l\}\Big) \\ &= \sum_{l=k}^{n} \binom{n}{l} F(x)^l (1 - F(x))^{n-l}. \end{aligned}$$

Hence

$$\boxed{F_{X^{(k)}}(x) = \sum_{l=k}^{n} \binom{n}{l} F(x)^l (1 - F(x))^{n-l}.} \tag{0.1}$$

One can see that the pdf $f_{X^{(k)}}$ exists and is given by (hint: differentiate $F_{X^{(k)}}$)

$$\begin{aligned}
f_{X^{(k)}}(x) &= \sum_{l=k}^{n} \binom{n}{l} l f(x) F(x)^{l-1} (1-F(x))^{n-l} \\
&\quad - \sum_{l=k}^{n} \binom{n}{l} (n-l) f(x) F(x)^{l} (1-F(x))^{n-l-1} \\
&= \sum_{l=k}^{n} \frac{n!}{(l-1)!(n-l)!} f(x) F(x)^{l-1} (1-F(x))^{n-l} \\
&\quad - \sum_{l=k}^{n} \frac{n!}{l(n-l-1)!} f(x) F(x)^{l} (1-F(x))^{n-l-1} \\
&= \sum_{l=k}^{n} \frac{n!}{(l-1)!(n-l)!} f(x) F(x)^{l-1} (1-F(x))^{n-l} \\
&\quad - \sum_{l=k+1}^{n} \frac{n!}{(l-1)(n-l)!} f(x) F(x)^{l-1} (1-F(x))^{n-l} \\
&= \frac{n!}{(k-1)!(n-k)!} f(x) F(x)^{k-1} (1-F(x))^{n-k}
\end{aligned}$$

i.e.

$$\boxed{f_{X^{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} f(x) F(x)^{k-1} (1-F(x))^{n-k}, x \in \mathbb{R}}$$

Distribution of $(X^{(1)}, X^{(n)})$:

Let

$$\mathbb{R}^2_{\leq} = \{(x,y) | x \leq y\}, \mathbb{R}^2_{<} = \{(x,y) | x < y\}.$$

Since $P\big((X^{(1)}, X^{(n)}) \in \mathbb{R}^2_{<}\big) = 1$, we compute $P\big(X^{(1)} \leq x, X^{(n)} \leq y\big)$ for $(x,y) \in \mathbb{R}^2_{\leq}$. Observe that

$$P\big(X^{(1)} \leq x, X^{(n)} \leq y\big) = P(X^{(n)} \leq y) - P\big(X^{(1)} > x, X^{(n)} \leq y\big).$$

$$\begin{aligned}
P\big(X^{(1)} > x, X^{(n)} \leq y\big) &= P\big(x < X_i \leq y, \text{ for all } i = 1, \cdots, n\big) \\
&= (F(y) - F(x))^n.
\end{aligned}$$

Therefore, for $(x,y) \in \mathbb{R}^2_{\leq}$

$$P\big(X^{(1)} \leq x, X^{(n)} \leq y\big) \;=\; F_{X^{(n)}}(y) - (F(y)-F(x))^n \;=\; F(y)^n - (F(y)-F(x))^n.$$

Observing that

$$\begin{aligned} f_{1n}(x,y) &= \frac{\partial^2}{\partial x \partial y} P\big(X^{(1)} \leq x, X^{(n)} \leq y\big) \\ P\big(X^{(1)} \leq x, X^{(n)} \leq y\big) &= P\big(X^{(1)} \leq x-\varepsilon, X^{(n)} \leq y\big),\ (x,y) \notin \mathbb{R}^2_{\leq} \end{aligned}$$

for $\varepsilon > 0$ small enough, we get

$$f_{1n}(x,y) \;=\; \begin{cases} n(n-1)f(x)(F(y)-F(x))^{n-2} & \text{if } (x,y) \in \mathbb{R}^2_{<}, \\ 0 & \text{otherwise.} \end{cases}$$

Example 0.2 *Let $X_1, \cdots, X_n$ be i.i.d. Uniform $(0,1)$ random variables. Compute the marginal distributions of the 2nd order statistics. Also find the joint distribution of $\min\{X_1, \cdots, X_n\}$ and $\max\{X_1, \cdots, X_n\}$. Hence find $P(\min\{X_1, \cdots, X_n\} + \frac{1}{2} > \max\{X_1, \cdots, X_n\})$*

Note the pdf f_k of the kth order statistics $X^{(k)}$ is given by

$$\begin{aligned} f_k(x) &= \frac{n!}{(k-1)!(n-k)!} f(x) F(x)^{k-1} (1-F(x))^{n-k}, x \in \mathbb{R} \\ &= \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k},\ 0 < x < 1,\ =\ 0, \text{otherwise.} \end{aligned}$$

i.e. $X^{(k)}$ has Beta $(k, n-k+1)$ distribution[2]*.*

The joint distribution of $X := \min\{X_1, \cdots, X_n\},\ Y := \max\{X_1, \cdots, X_n\}$ is given by

$$\begin{aligned} f_{1n}(x,y) &= n(n-1)f(x)f(y)(F(y)-F(x))^{n-2},\ x < y, \\ &\quad =\ 0,\ \text{otherwise} \\ &= n(n-1)(y-x)^{n-2},\ 0 < x < y < 1,\ =\ 0,\ \text{otherwise.} \end{aligned}$$

[2]Beta (n,m) random variable is defined through the pdf

$$f(x) \;=\; \begin{cases} \frac{(n+m-1)!}{(m-1)!(n-1)!} x^{m-1}(1-x)^{n-1}, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Hence the required probability is given by

$$\begin{aligned} P(Y < X + \frac{1}{2}) &= n(n-1)\int_0^{\frac{1}{2}}\int_{x_1}^{x+\frac{1}{2}}(y-x)^{n-2}dydx \\ &\quad +n(n-1)\int_{\frac{1}{2}}^{1}\int_{x}^{1}(y-x)^{n-2}dydx \\ &= n(n-1)\Big[\frac{1}{2^n(n-1)}+\frac{1}{2^n n(n-1)}\Big] = \frac{n+1}{2^n}. \end{aligned}$$

Deriving Laplace distribution: Given a data $\{x_1, x_2, \cdots, x_n\}$, find the distribution of the error (true value minus observation) given the postulate: median is the most probable statistic for the true value.

We assume that error is with pdf. So probability that all data points are distinct is 1 and we discard a data set if x_i's are not distinct (*).
Also data is collected independently.

Consider the likelihood function for sample $\{x_1, x_2, \cdots, x_n\}$ is

$$L(x; x_1, x_2, \cdots, x_n) \;=\; \varphi(x-x_1)\cdots\varphi(x-x_n)$$

where φ is the pdf of the error. Now the postulate is most probable value is the median say x_{med}. Set

$$\psi(x) \;=\; \frac{\varphi'(x)}{\varphi(x)},\; x \in \mathbb{R}.$$

Then using $\varphi(x) = \varphi(-x)$, we get

$$-\psi(-x) \;=\; \psi(x),\; x \in \mathbb{R} \qquad (1)$$

Using the postulate, we get

$$\sum_{k=1}^{n}\psi(x_k - x_{med}) \;=\; 0 \qquad (2)$$

Order the data $x^{(1)} < x^{(2)} < \cdots < x^{(n)}$.
When $n = 1$, then $x_{med} = x_1$ and hence using (2), we get

$$\psi(0) \;=\; 0. \qquad (3)$$

For $n = 3$, $x_{med} = x^{(2)}$ and using (2) we get

$$\begin{aligned} \psi(x^{(1)} - x^{(2)}) + \psi(0) + \psi(x^{(3)} - x^{(2)}) &= 0 \\ &\Rightarrow \psi(x^{(3)} - x^{(2)}) = \psi(x^{(2)} - x^{(1)}). \end{aligned}$$

For the last equality, we use (1). Since the above is true for all samples satisfying (*), we get

$$\psi(\theta_1) = \psi(\theta_2),\ \theta_1, \theta_2 > 0. \qquad (4)$$

i.e.

$$\psi(x) = c,\ x > 0.$$

Now using (1), we get

$$\psi(x) = -c\ x < 0.$$

Therefore

$$\psi(x) = c\,\mathrm{sign}(x),\ x \in \mathbb{R}.$$

Hence

$$\frac{d}{dx} \ln \varphi(x) = c\,\mathrm{sign}(x).$$

Therefore,

$$\varphi(x) = ke^{c|x|}, x \in \mathbb{R}.$$

Since φ is pdf, $c < 0$. Hence

$$\varphi(x) = \frac{\alpha}{2} e^{-\alpha|x|}, x \in \mathbb{R},\ \alpha > 0.$$

i.e. the postulate implies the error distribution is given by a Laplace distribution.

Repeat the same calculation with error distribution as Laplace distribution, we can see that most probable value for the true value given a data set is its median.

Chapter 7 : Expectation

Key words: Expectation of discrete random variable, simple random variable, expectation of non negative random variable, expectation of general random variable, monotone convergence theorem, dominated convergence theorem.

In this chapter, we introduce expected value or the mean of a random variable. First we define expectation for discrete random variables and then for general random variable.

First we give a useful representation of discrete random variables.

Theorem 0.3 *Let X be a discrete random variable defined on a probability space $(\Omega, \mathcal{F}, P)$. Then there exists a partition $\{A_n \mid n = 1, , \ldots, N\} \subseteq \mathcal{F}$ of Ω and $\{a_n \mid n = 1, 2, \ldots, N\} \subseteq \mathbb{R}$ with a_n's distinct, so that*

$$X \;=\; \sum_{n=1}^{N} a_n I_{A_n} \text{ a.s.},$$

where N may be ∞.

Proof. Let F be the distribution function of X. Let $\{a_2, \ldots, a_N\}$ be the set of all discontinuities of F. Here N may be ∞. Since X is discrete, we have

$$\sum_{n=2}^{N} F(a_n) - F(a_n-) \;=\; 1\,.$$

Set

$$A_n \;=\; \{X = a_n\}\,.$$

Then $\{A_n\}$ is pairwise disjoint. Let $\Omega' = \cup_{n=1}^{N} A_n$, a_1 be such that $a_1 \neq a_n$ for all $n \geq 2$ and $A_1 = \Omega \setminus \Omega'$. Then $\{A_n : n \geq 1\}$, is a partition of Ω, $P(A_1) = 0$ and

$$\Big\{X \neq \sum_{n=1}^{N} a_n I_{A_n}\Big\} \subseteq A_1.$$

Hence $P\Big(X \neq \sum_{n=1}^{N} a_n I_{A_n}\Big) = 0$, i.e.

$$X \;=\; \sum_{n=1}^{N} a_n I_{A_n} \text{ a.s.},$$

Remark 0.2 *(i) If X is a discrete random variable on a probability space $(\Omega, \mathcal{F}, P)$, then the 'effective' range of X is at the most countable. Here 'effective' range means those values taken by X which has positive probability. This leads to the name 'discrete' random variable.*

(ii) If two random varibles are such that $X = Y$ a.s., then $F_X(x) = F_Y(x)$ for all x (exercise).

Remark 0.3 *If X is a discrete random variable given by $(A_n, a_n)|n = 1, 2, \cdots, N\}$ then one can assume without the loss of generality that*

$$X \;=\; \sum_{n=1}^{\infty} a_n I_{A_n}\,.$$

Since if $N < \infty$, then set $A_n = \emptyset$ for $n \geq N+1$ and $a_n, n \geq N+1$ are chosen so that they are distinct.

Theorem 0.4 *Let $\{A_n\}$ and $\{B_n\}$ be countable partitions of Ω from $\mathcal{F}$ and $\{a_n\}$, $\{b_n\}$ be a sequence of real numbers such that a_n's are distinct and b_n's are not necessarily distinct. Then if*

$$\sum_{n=1}^{\infty} a_n I_{A_n} \;=\; \sum_{n=1}^{\infty} b_n I_{B_n}$$

then

$$\sum_{n=1}^{\infty} a_n P(A_n) \;=\; \sum_{n=1}^{\infty} b_n P(B_n)$$

provided l.h.s. series converges absolutely.